

Operations Research: theory and applications to networking

Covering with capacity constraints

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ICT4SS-CE

Covering with capacity constraints

Given:

- a set of **potential facility locations** (e.g., warehouses, service centers, base stations),
- a set of **demand points** (e.g., customers, cities, users),
 - for each facility:
 - a **fixed opening cost**,
 - a **maximum capacity** (how much demand it can serve),
 - for each pair (facility, demand point):
 - a **service cost** (increasing with the distance),
 - for each demand point:
 - a **demand quantity**.

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Objective:

Decide

- **which facilities to open**, and
- **how to assign demand points to open facilities**,

so that:

- **every demand point is covered (served)**,
- **no facility exceeds its capacity**,
- **the total cost** (facility opening + service/assignment cost) is minimized

Find a ILP fomulation of the problem

Properties of the problem

- Does the problem always admits a solution?
Which are minimal conditions that guarantee the problem to have a solution?
- Build arbitrarily a few non trivial instances of the problem and solve them with Xpress.